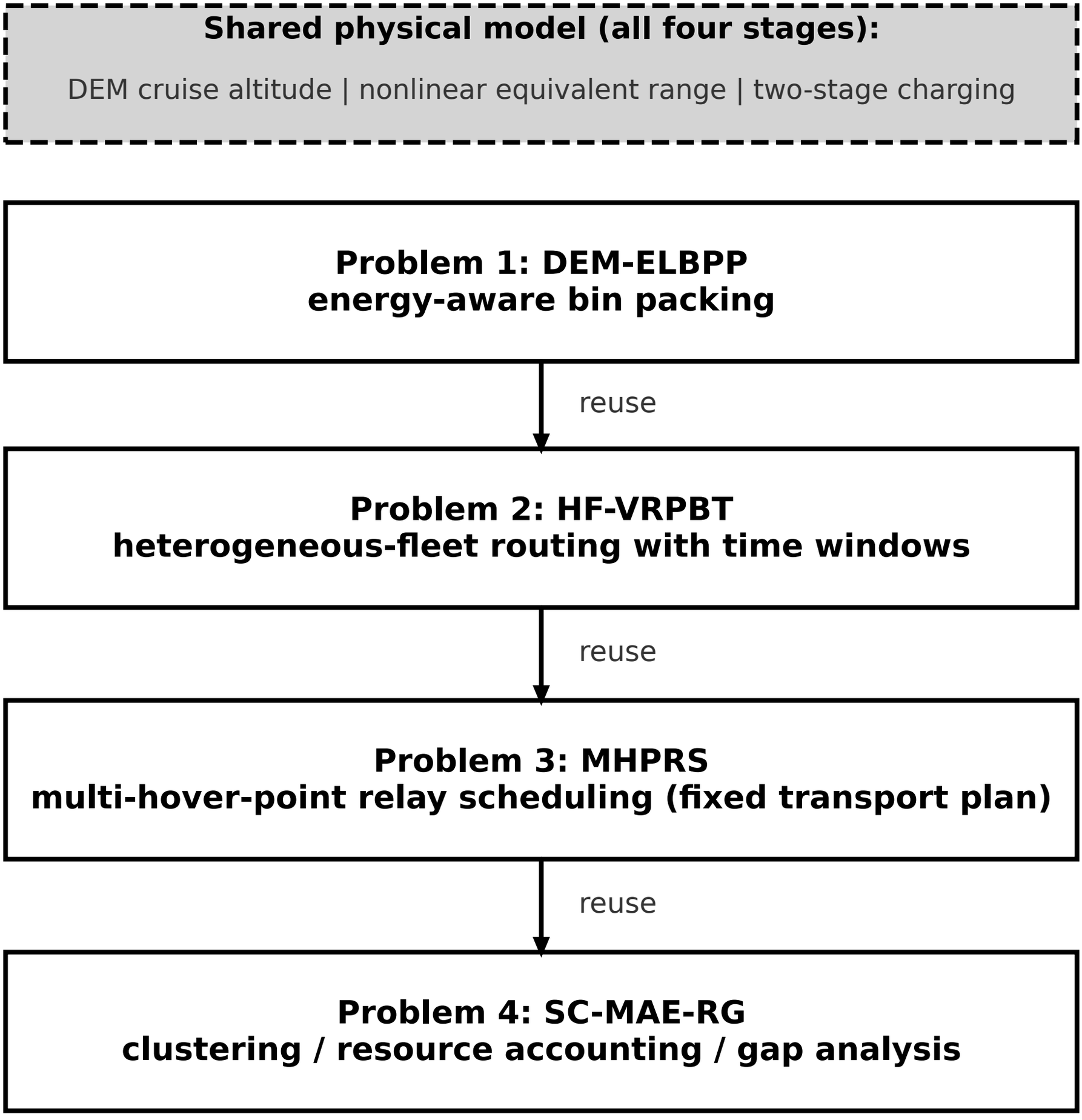
Shared physical model (all four stages):
DEM cruise altitude | nonlinear equivalent range | two-stage charging
Problem 1: DEM-ELBPP
energy-aware bin packing
reuse
Problem 2: HF-VRPBT
heterogeneous-fleet routing with time windows
reuse
Problem 3: MHPRS
multi-hover-point relay scheduling (fixed transport plan)
reuse
Problem 4: SC-MAE-RG
clustering / resource accounting / gap analysis